Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – VII (Batch 2024-2028)

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 7	PRO	24INT451	Summer Internship	0	0	0	0	2
	OE	24BT421T	Biosimilars Technology	3	0	0	3	3
	PE	24BT431T	Pharmaceutical Biotechnology	3	0	0	3	3
		24BT432T	Synthetic and systems biology					
		24BT433T	Metabolic engineering					
	PE	24BT434T	Biomass conversion and Utilisation	3	0	0	3	3
		24BT435T	Stem Cell Technology					
		24BT436T	Molecular Data Analysis and Simulations					
	PC	24BT401T	Bioethics, IPR and Biosafety	4	0	0	4	4
	PC	24BT402T	Nanotechnology	3	0	0	3	3
	PC	24BT402P	Nanotechnology Laboratory	0	0	2	2	1
	PC	24BT403T	Downstream Processing	3	0	0	3	3
	PC	24BT403P	Downstream Processing Laboratory	0	0	2	2	1
	PRO	24PRBT451	Seminar					1

24INT451					Summer Internship					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	2	-	-	-	-	-	-	100

COURSE OBJECTIVES

1. Provide students with a comprehensive understanding of industrial biotechnology, including biopharmaceutical manufacturing, fermentation technology, nanotechnology, and targeted drug delivery, emphasizing real-world applications.
2. Expose students to emerging trends and challenges in vaccine development, antibody engineering, and medical diagnostics, fostering an appreciation for regulatory compliance, quality assurance, and preclinical validation.
3. Enhance technical expertise in computational biology and AI-driven drug discovery, enabling students to explore bioinformatics, molecular docking, and machine learning approaches for precision medicine.
4. Develop problem-solving and research skills through exposure to environmental biotechnology, food and agricultural biotechnology, and sustainable bio-manufacturing, promoting innovation in bioremediation, biofuels, and functional foods.

Suggested field of exposure is summarised below but not limited to the mentioned theme of Industrial orientation:

1. Biopharmaceuticals & Drug Manufacturing
   - GMP (Good Manufacturing Practices) in Biopharma
   - Upstream & Downstream Bioprocessing
   - Quality Control & Assurance in Drug Production
   - Monoclonal Antibody Production & Purification
   - Recombinant Protein Expression & Therapeutic Development

2. Fermentation Technology & Industrial Enzymes
   - Large-Scale Fermentation & Bioreactor Operations
   - Enzyme Production & Optimization for Industrial Applications
   - Microbial Biotechnology for Sustainable Bio-Manufacturing

3. Nanotechnology & Targeted Drug Delivery
   - Nanocarriers and Liposomes for Drug Delivery
   - Polymer-Based Scaffolds for Tissue Engineering
   - Nano-Encapsulation for Controlled Drug Release
   - Nanosensors for Disease Detection & Monitoring

4. Food & Agricultural Biotechnology
   - Probiotics, Prebiotics & Functional Foods
   - Plant Tissue Culture & Genetic Modification Techniques
   - Biotechnology in Crop Improvement & Stress Resistance
   - Microbial Biotechnology for Soil Health & Biofertilizers.

5. Environmental & Industrial Biotechnology
   - Bioremediation & Wastewater Treatment Technologies
   - Biofuel Production & Sustainable Biotechnology

- Microbial Fuel Cells & Bioconversion Processes
- Biosensors for Environmental Monitoring & Pollution Control

6. Vaccine Development & Immunobiotechnology
   - Vaccine Formulation & Antigen Development
   - mRNA-Based Vaccines & Future Innovations
   - Clinical & Preclinical Validation of Vaccines
   - Regulatory & Quality Control Aspects in Vaccine Production

7. Medical Biotechnology & Diagnostics
   - Molecular Diagnostics & Biomarker Discovery
   - PCR, ELISA, and Next-Generation Sequencing in Diagnostics
   - Point-of-Care Diagnostic Technologies
   - Stem Cell Research & Regenerative Medicine

8. Computational Biology & AI in Drug Discovery
   - Machine Learning for Drug Target Identification
   - In-Silico Screening & Molecular Docking Simulations
   - Bioinformatics & Systems Biology for Disease Modeling
   - Computational Approaches in Precision Medicine

9. Antibody Engineering & Biologics
   - Hybridoma Technology for Monoclonal Antibody Production
   - Recombinant Antibody Engineering & Therapeutics
   - Biosimilar Development & Biopharmaceutical Regulations

10. Pharmaceutical Research & Preclinical Validation
    - Preclinical Drug Screening & Toxicology Studies
    - Pharmacokinetics & Pharmacodynamics Analysis
    - Regulatory Compliance & Clinical Trials
    - Personalized Medicine & Drug Repurposing Strategies

Outcomes will be assessed on a scale of 100 marks.

Presentation & Screening Format:

- Each student will present before a panel of faculty and industry representatives.
- **Evaluation Criteria for Presentation:**
  - Clarity of content and technical knowledge
  - Ability to articulate industrial learning
  - Handling of Q&A session by the panel
  - Professionalism in communication

Assessment Criteria's

The assessment of the Industry Orientation program will be based on four key components. Industry attendance and supervisor feedback will evaluate student engagement, punctuality, and teamwork as observed by industry mentors. Project report submission will assess the detailed documentation of industry work, methodologies, and learning outcomes. Technical presentation and viva will measure the student's ability to present key findings before a faculty and industry expert panel. Lastly, professional skills and committee review will focus on critical thinking, problem-solving, and communication skills demonstrated during discussions with the evaluation panel.

COURSE OUTCOMES

On completion of the Industrial orientation, students will be able to

CO1: Demonstrate an understanding of biopharmaceutical manufacturing processes, including GMP standards, monoclonal antibody production, recombinant protein expression, and quality control in drug development.

CO2: Analyze the principles and applications of fermentation technology and industrial enzyme production, focusing on large-scale bioprocessing, microbial biotechnology, and process optimization.

CO3: Evaluate nanotechnology-based drug delivery systems, such as nanocarriers, polymer scaffolds, and nanosensors, for biomedical and pharmaceutical applications.

CO4: Apply knowledge of food, agricultural, and environmental biotechnology to real-world challenges, including plant tissue culture, probiotic development, bioremediation, and biofuel production.

CO5: Investigate the latest advancements in vaccine development, immunobiotechnology, and medical diagnostics, understanding their role in disease prevention, molecular diagnostics, and regenerative medicine.

CO6: Utilize computational biology and AI tools for drug discovery, molecular docking, bioinformatics analysis, and personalized medicine applications, integrating technology with biotechnology research.

24BT421T					BIOSIMILARS TECHNOLOGY					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand how a protein's structure dictates its function.
2. To compare the different host-expression systems as vectors required for biosimilar production
3. To study the different steps and techniques involved in processing of biosimilar
4. To categorize the levels and guidelines of clinical trials

UNIT I: INTRODUCTION 10 Hrs.

Introduction, protein and its broad functions, outline of protein synthesis and its importance, post-translation modifications, therapeutic proteins. Introduction to biosimilars, definition, examples, uses and applications. Overview of Pharmaceutical Biotechnology and Biosimilars technology

UNIT II: HOST EXPRESSION SYSTEMS 10 Hrs.

Biologics/Biosimilars: A Regulatory Overview, Different expression hosts with history and genotypes Prokaryotes - E.coli DH5 alpha , E.coli BL21A1, E.coli BL21DE3 etc. Eukaryotes – Yeast hosts like Pichia pastoris , Mammalian hosts like CHOK1, CHO DuxB11, CHO DG44, NS0, SP02 cell line

UNIT III: PROCESSING OF BIOSIMILARS 10 Hrs.

Primary processing using microbial cultures, Microbial and Mammalian protein purification, cell separation, cell disruption, protein isolation, protein purification processes – chromatography and ultrafiltration. Formulation of biosimilar drugs, stability studies pre-packaging and post packaging of biosimilar drugs. Types and source of impurities in biosimilars and method of identifying impurities.

UNIT IV: CLINICAL TRIALS 10 Hrs.

Regulatory Compliance for Biopharmaceuticals and Other Biologics, Concepts of non-clinical animal trials and clinical trials on human volunteers (Phase I, II, III, IV clinical trials), Guidelines and Case studies

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Explain the different functions of proteins, therapeutic proteins
CO2 : Understand the different course expression systems in prokaryotes and eukaryotes involved in the production of biosimilars
CO3 : Apply the methods of Pharmaceutical technology for the production of biosimilars
CO4 : Classify the different vector systems used in biosimilar production
CO5 : Appraise the production and processing of biosimilars and therapeutics
CO6 : Formulate the animal and clinical trials for the testing and validation of therapeutics

TEXT/REFERENCE BOOKS

1. Grindley, Jill E. Ogden. Understanding Biopharmaceuticals: Manufacturing and Regulatory Issues. CRC Press
2. Crommelin D.J.A. , Sindelar R. D ,Bernd Meibohm. Pharmaceutical Biotechnology, II edition, Springer
3. Gary Walsh. Pharmaceutical Biotechnology. Wiley
4. O. Kayser, R. H. Muller Pharmaceutical Biotechnology. Wiley - VCH
5. Jay P Rho, Stan G Louie. Handbook of Pharmaceutical Biotechnology. Haworth Press.

24BT431T					Pharmaceutical Biotechnology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understanding the importance pharmaceutical products.
2. To develop the skills necessary for employing pharmaceutical biotechnology principles.
3. To learn the significance of pharmaceutical preparation and method of production of Vaccines and protein based drugs
4. To evaluate the different regulations and strategy involved in the current and future biotechnology related products on the market.

UNIT I: PHARMACEUTICAL PREPARATIONS 12Hrs

Classification of Dosage forms I. Solid Dosage: Powders, Tablets, Capsules, and Granules. II. Semi solid Dosage: Creams, Gels, Ointment and Paste. III. Liquid Dosage: Monophasic (Syrups & Elixirs), Biphasic liquids: Suspension, Emulsion. IV. Gas Dosage: Aerosols. Different routes of drug administration: Oral, Parenteral, Dermal, Nasal, Ocular, Rectal and their merits & demerits. Types of impurities in pharmaceuticals, Analytical Techniques for Impurity Profiling, Role of stability studies in impurity profiling.

UNIT II: VACCINES 10Hrs

General method of the preparation of bacterial vaccines, viral vaccine, antitoxins. Screening of vaccines in animal models.

UNIT III: PHARMACOLOGY OF PROTEIN DRUGS 10 Hrs

Types of immunity and hypersensitivity reactions, Classification and Pharmacology of Immunosuppressant including Glucocorticoids, Calcineurin inhibitors, m-TOR inhibitors, IL- receptor antagonist and TNF inhibitors.

UNIT IV: PHARMACEUTICAL PRODUCT DEVELOPMENT 10Hrs

Biosensors-Types, Working and applications of biosensors in Pharmaceutical Industries. Pre-clinical studies: (Safety and Efficacy), Clinical studies (Phase I-IV), CDSCO- Regulatory requirements and approval, Drug distribution cycle.

Max.42 Hr.

COURSE OUTCOMES

On completion of the course, student will be able to;

CO1: Demonstrate the challenges in the development of pharmaceutical products

CO2: Evaluate the biotechnology based pharmaceutical products and its screening methods

CO3: Analyze the scope and opportunities of different dosage forms and protein based drugs in treating different medical conditions in healthcare sectors.

CO4: Build the biotechnology applications in the pharmaceutical field

CO5: Illustrate the pharmacological mechanisms of protein based drugs

CO6: Develop and validate newer formulation and protein based drugs using modern biotechnology concepts

TEXT/REFERENCE BOOKS

1. R. S. Satoskar, S. D. Bhandarkar, Nirmala N. Rege," Pharmacology and pharmacotherapeutics", Popular Prakashan P. Ltd.- New Delhi.
2. Rang H. P., Dale M. M., Ritter J. M., Flower R. J, "Rang and Dale's Pharmacology",Churchil Livingstone Elsevier.
3. Katzung B. G., Masters S. B., Trevor A. J,"Basic and clinical pharmacology", Tata Mc Graw-Hill.

4. K.D.Tripathi, “Essentials of Medical Pharmacology”, JAYPEE Brothers Medical Publishers (P) Ltd, New Delhi.
5. Ashok Mulchandani and Kim Rogers, “Enzyme and Microbial Biosensors: Techniques and Protocols”,Springer.
6. Bjorn Knollmann, “Goodman and Gilman’s, The Pharmacological Basis of Therapeutics”, McGraw-Hill Education.
7. Sharma H. L., Sharma K. K., “Principles of Pharmacology”, Paras medical publisher.
8. Lachman Liebermans., “The Theory and Practice of Industrial Pharmacy”, 4th Edition, CBS publisher.
9. Loyd.V.Allen., “ Ansel’s Pharmaceutical Dosage Form and Drug Delivery System”, 11th Edition, Wolters Kluwer India Pvt. Ltd publisher.
10. Parthasarathi G, Karin Nyfort-Hansen, Milap C Nahata, “A textbook of Clinical Pharmacy Practice-essential concepts and skills, Orient Longman Private Limited publisher.

24BT432T					Synthetic and Systems Biology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Learn the specific information of synthetic circuits
2. Gain the fundamental on the various models for modelling biological circuit.
3. Create interest in learning the tools and techniques used in network expression data analysis.
4. Learn the applications and scopes of genomics and proteomics research.

UNIT I: INTRODUCTION 10 Hrs.

Transcriptional and Post-transcriptional regulation: Biology & engineering systems, modeling foundation: Chemical kinetics (M-M equation and Hill function), Deterministic models, And Stochastic models Spatiotemporal models. Noise in gene expression: Origin, propagation, consequences, and control, Robustness and evaluability of genetic networks, Bacterial circuits.

UNIT II: SYNTHETIC CIRCUITS 10 Hrs.

Genetic Circuits in Eukaryotes: How to build genetic circuits in yeast Logic gates in yeast Scale up to mammalian cells, Introduction to non-coding RNAs, ncRNAs Characteristics, biogenesis, and activities in cell physiology and developmental process, Application of ncRNAs on circuit designs Riboswitches ; microRNAs ; siRNAs.

UNIT III: NETWORK BIOLOGY AND APPLICATIONS 10 Hrs.

Network biology and human diseases; Signal Transduction and Gene Networks; Protein–protein interaction networks. Gene regulatory networks (DNA–protein interaction networks); Gene co-expression networks (transcript–transcript association networks).

UNIT IV: METHOD AND SOFTWARE PLATFORM FOR SYSTEMS BIOLOGY 10 Hrs.

SBML models and MATHSBML, Cell designer, Systematic detection of Biological Networks, Storing, Searching and Dissecting experimental proteomic data.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify the different biological circuits.
CO2 : Gain the fundamental practice of non-coding RNAs.
CO3 : Understand about the pipelines for cell based circuit designs and riboswitches.
CO4 : List different genetic networks and expression networks.
CO5 : Focus professionally on methods and platforms for synthetic and systems biology
CO6 : Develop solution to enhance understanding for biological circuit and expression networks.

TEXT/REFERENCE BOOKS

1. Dokholyan, Nikolay, “Computational Modeling of Biological Systems: From Molecules to Pathways” Springer-Verlag New York, 2019
2. Klipp E Wolfran L, “System Biology: A Text Book” Wiley-VH Verlag Gmbh ,2018
3. Alon Uri, “Introduction to Systems Biology: Design Principles of Biology Circuits” Chapman & Hall/CRC/, 2020
4. JM Bower and H Bolouri, eds, “Computational modeling of genetic and biochemical networks”, MIT Press 2018

5. Fall, C.P., Marland, E.S., Wagner, J.M., Tyson, J.J, **"Computational Cell Biology",** Springer, 2018.
6. Choi Sangdun, **"Introduction to System Biology",** Humana Press/Trtowa/New Jersey, 2018.

24BT433T	Metabolic Engineering

Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To analyze mechanisms of Metabolic Control
2. To evaluate strategies for enhancing Primary Metabolite Production
3. To understand the regulation of Secondary Metabolite Biosynthesis
4. To design and analyze bioconversion Strategies

UNIT I: INTRODUCTION 12 Hrs.

Induction-Jacob Monod Model, catabolite regulation, glucose effect, camp deficiency, feedback regulation, regulation in branched pathways, differential regulation by isoenzymes, concerted feedback regulation, cumulative feedback regulation, amino acid regulation of RNA synthesis, energy charge, permeability control passive diffusion, facilitated diffusion, active transport group transportation

UNIT II: SYNTHESIS OF PRIMARY METABOLITES 10 Hrs.

Alteration of feedback regulation, limiting accumulation of end products, feedback, resistant mutants, alteration of permeability.

UNIT III: BIOSYNTHESIS OF SECONDARY METABOLITES 10 Hrs.

Precursor effects, prophophase, idiophase relationships, enzyme induction, feedback regulation, catabolite regulation by passing control of secondary metabolism, producers of secondary metabolites.

UNIT IV: BIOCONVERSIONS 10 Hrs.

Advantages of Bioconversions, specificity, yields, factors important to bioconversions, regulation of enzyme synthesis, mutation, permeability, co-metabolism, avoidance of product inhibition, mixed or sequential bioconversions, conversion of insoluble substances.

OTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Explain the Mechanisms of Gene Regulation
CO2 : Analyze the Regulation of Metabolic Pathways
CO3 : Strategies for Manipulating Primary Metabolite Synthesis
CO4 : Understand the Biosynthesis of Secondary Metabolites
CO5 : Analyze the Principles of Bioconversions
CO6 : Apply Knowledge to Design Bioconversion Strategies

TEXT/REFERENCE BOOKS

1. Jens Nielsen, Gregory Stephanopoulos, Sang Yup Lee, “Metabolic Engineering: Concepts and Applications”, Wiley , 2021
2. Wang D. I. C., Cooney C. L., Demain A. L., Dunnil P., Humphrey A. E., Lilly M. D., “Microbial Fermentation and Enzyme Technology”, John Wiles and Sons., 2020.
3. Stanbury P. F. and Whitaker A., “Principles of Fermentation Technology”, Pergamon Press, 2016.
4. Palmer T., “Enzymes”, Horwood Publishers, 2001

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24BT434T					Biomass Conversion and Utilization					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the pigmentary property, chemistry behind the colorants.
2. To differentiate the various biomass conversions procedures
3. To familiarize the properties of biomass and its energy products
4. To apply the concepts in the production and utilisation of biodiesel

UNIT I: BIOMASS – ENERGY SOURCE 10 Hrs.

Introduction: Types of biomass, advantages and dis advantages in use of biomass as energy, sources of biomass, current biomass applications and trends, Biological – Thermal – Chemical – Hybrid conversions – Application of biomass conversion.

UNIT II: THERMAL AND CHEMICAL CONVERSION 10 Hrs.

Thermo chemical conversion: Combustion, gasification, pyrolysis, hydrothermal liquefaction, hydropyrolysis, torrefaction, choice of thermal process based on biomass type and product requirement. Economics of thermo chemical conversion.

UNIT III: BIOLOGICAL CONVERSION 10 Hrs.

Biodegradation and biodegradability of substrate - Biochemistry and process parameters of biomethanation - Biogas digester types - Digester design and biogas utilization. Biomethanation Process - Economics of biogas plant with their environmental and social impacts - Bioconversion of substrates into alcohol - Methanol & ethanol Production - Organic acids – Solvents - Amino acids - Antibiotics etc

UNIT IV: BIODIESEL AND BIOMASS GASIFICATION 10 Hrs.

Biomass gasification – chemistry – types of gasifiers – gasifier design : TDR, throughput, A/F ratio and equivalence ratio calculations – advanced gasification – fluidized bed gasifier – component design – cold fluidization tests – Electrical power production – Biomass combustion – types of combustors – Co-combustion and Co-firing – applications – Eutectic point of biomass ash. Chemical Conversion: Hydrolysis & hydrogenation - Solvent extraction of hydrocarbons - Solvolysis of wood - Biocrude and biodiesel - Chemicals from biomass

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Demonstrate different types of biomass, advantages and limitations of bio mass,
CO2 : Explain thermo chemical conversion processes of biomass.
CO3 : Describe various biological conversion processes of bio mass.
CO4 : Discuss about chemical conversion processes of bio mass.
CO5 : Explain concepts of power generation using biomass.
CO6 : Apply the biomass assessment techniques.

TEXT/REFERENCE BOOKS

1. P. Wankat. “Separation Process Engineering”. Fifth Edition. 2022
2. DB Pal, P. Singh “Utilization of Waste Biomass in Energy”, Environment and Catalysis, CRC Press, 2022
3. NP Nghiem, TH Kim, CG Yoo. “Biomass utilization Conversion Strategies”, Springer, 2023.

4. Erik Dahlquist, “Biomass as Energy Source: Resources, systems and applications”, Sustainable Energy Developments series, 2013, CRC Press, Taylor and Francis Group.
5. D.P.Kothari, K.C Singal and Rakesh Ranjan “Renewable Energy Sources And Emerging Technologies”, 2011, PHI Learning Private Ltd, New Delhi.

24BT435T					Stem Cell Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/VIVA	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

1. To impart knowledge of wide-ranging topics related to stem cells and regenerative biology, including a brief history of the field
2. To impart, research on animal models of regeneration, tissue engineering, social and ethical issues related to stem cell research.
3. To Explore the Interdisciplinary Nature of Stem Cell Research
4. To Foster Critical Analysis of Stem Cell Research Methods and Applications

UNIT I: INTRODUCTION TO STEM CELLS AND STEM CELL NICHE 10 Hrs.

Principles and properties of stem cells, types of stem cells, comparison of embryonic and adult stem cells. Introduction to stem cell niches in gut epithelium, bone marrow, epidermis, testis, and neural tissues

UNIT II: CELL CYCLE AND DEVELOPMENT & EPIGENETIC CONTROL 10 Hrs.

Cell cycle regulators and checkpoints, cell fusion, differentiation of stem cells and their role in self-renewal DNA-methylation and histone modifications, genomic imprinting, telomerase regulation, X-chromosome inactivation, reprogramming of cells.

UNIT III: TYPES AND REGENERATION 10 Hrs.

Early asymmetric division and generation of symmetry in developing embryo in animals and plants; organogenesis and morphogenesis, metamorphosis, animal life cycle, sex determination and role of apoptosis in organ development; Role of morphogens and their gradient in axis patterning and determination. Concept of anteroposterior, dorso-ventral, and medio-lateral axis formation.

UNIT IV: THERAPEUTIC APPLICATIONS OF STEM CELLS AND ETHICAL ISSUES 10 Hrs.

Gene Therapy: Introduction, History and evolution of Gene therapy, optimal disease targets, Failures and successes with gene therapy and future prospects, Genetic Perspectives for Gene Therapy, Gene Delivery methods: Viral vectors and Non-viral Vectors, Regulatory and Ethical Considerations of stem cell and Gene Therapy, Assessing Human Stem Cell Safety, Use of Genetically Modified Stem Cells in Experimental Gene Therapies.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain fundamental knowledge on developmental biology of plant and animals
CO2 : Understand the biology of Cell cycle regulators
CO3 : Classify and compare the animals and plants; organogenesis and morphogenesis
CO4 : Get acquainted with techniques on Stem Cell isolation
CO5 : Focus on applications of stem cells
CO6 : Design construct the recombinant vector and develop genetically modified organisms

TEXT/REFERENCE BOOKS

1. Treleaven, J," Hematopoietic Stem Cell Transplantation", first edition 2009.
2. Lanza, R," Essentials of Stem Cell Biology" second Edition, 2009 Academic Press.
3. Lodish et al, Molecular Cell Biology by sixth Ed., W.H. Freeman & Co. 2021.
4. Alexander Battler, Jonathan Leo, "Stem Cell and Gene-Based Therapy: Frontiers in Regenerative Medicine", Springer. 2006.

24BT436T					Molecular Data Analysis and Simulations					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Learn the critical relationship among bimolecular structure, function and force field models.
2. Gain the fundamental on drug/receptor interactions using in silico techniques.
3. Create interest in learning molecular mechanics, molecular dynamics simulations and homology modeling.
4. Understand the basic modeling techniques to explore biological phenomena at the molecular level.
5. Learn the concepts of energy minimisation and force fields.

UNIT I: INTRODUCTION 10 Hrs.

Useful Concepts in Molecular Modelling, Coordinate Systems, Internal Coordinates, Potential Energy Surfaces, Molecular Graphics, Surfaces, Computer Hardware and Software, The Molecular Modelling Literature.

UNIT II: FORCE FIELDS 10 Hrs.

Force Fields, Bond Stretching, Angle Bending, Torsional Terms, Introduction to Non-bonded Interactions: Electrostatic Interactions; Vander Waals Interactions, Hydrogen Bonding in Molecular Mechanics, Force Field Models for the Simulation of Liquid Water.

UNIT III: MOLECULAR DYNAMICS AND MONTE CARLO SIMULATION 10 Hrs.

Molecular dynamics simulation methods, Constant temperature and pressure, Monte Carlo simulation of molecules, Models used in Monte Carlo simulations of polymers, Molecular modeling software: BIOSUITE and AMBER.

UNIT IV: ENERGY MINIMISATION AND COMPUTER SIMULATION 10 Hrs.

Energy minimization and related methods for exploring the energy surface, Non-Derivative methods and Derivative methods, 1st and 2nd order minimisation methods, Computer simulation methods, Simple thermodynamic properties and phase space, Boundaries, Analyzing the results of a simulation and estimating errors, GROMACS and DESMOND.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify the computer Aided Drug Design methods and predict drug like compounds.
CO2 : Gain the fundamental practice of Molecular modeling and modern drug discovery.
CO3 : Understand about the rational drug design, based on three-dimensional structures and physicochemical properties of drugs and receptors.
CO4 : List different force fields and their optimization for drug design and simulations.
CO5 : Focus professionally on energy minimization and computer simulation
CO6 : Develop solution to enhance understanding for drug target interactions.

TEXT/REFERENCE BOOKS

1. Leach A.R. 2019, “Molecular Modelling Principles and Application”, 4th Edition. Pearson – Prentice Hall, New Delhi.
2. Satya P.G. 2020, “QSAR and Molecular Modeling”, Springer - Anamaya Pub, New Delhi.
3. Claude N. C. 2019. “Molecular Modeling in Drug Design”, Academic Press, California.
4. Kunal R. 2018, “Advances in QSAR Modeling: Applications in Pharmaceutical, Chemical, Food, Agricultural and Environmental Sciences”, Springer. ISBN 978-3-319-56849-2.

24BT401T					Bioethics, IPR and Biosafety					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
4	0	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To disseminate knowledge on patents, patent regime in India and abroad and registration aspects
2. To make students aware about current trends in IPR and Govt. supports in promoting IPR
3. To classify the role of regulatory committees in controlling the risk.
4. To Explore the Role of Intellectual Property in Innovation and Entrepreneurship

UNIT I: INTELLECTUAL PROPERTY RIGHTS 10 Hrs.

Introduction and the need for intellectual property right (IPR) - Kinds of Intellectual Property Rights: Patent, Copyright, Trade Mark, Design, Geographical Indication, Plant Varieties and Layout Design – Genetic Resources and Traditional Knowledge – Trade Secret - IPR in India: Genesis and development – IPR in abroad - Major International Instruments concerning Intellectual Property Rights. Plagiarism; Citation and acknowledgement

UNIT II: PATENT RIGHTS 10 Hrs.

Origin, Meaning of Patent, Types, Inventions which are not patentable, Registration Procedure, Rights and Duties of Patentee, Assignment and license. DESIGN- Meaning, Definition, Object, Registration of Design. TRADE MARKS— Origin, Meaning & Nature of Trade Marks, Types, Registration of Trade Marks.

UNIT III: BIOETHICS 10 Hrs.

Institutional Biosafety Committee, Institutional Animal ethics committee, Institutional review board, Human Genome project and ethical issues, Animal cloning, human cloning and their ethical issues, Experimentation on animals, Hazardous materials used in biotechnology, their handling and disposal.

UNIT IV: BIOSAFETY 10 Hrs.

Primary Containment for Biohazards; Biosafety Levels; Biosafety Levels of Specific Microorganisms; Recommended Biosafety Levels for Infectious Agents and Infected Animals; Safety in genetic and tissue engineering research; GMOs & LMOs; protocol of drug administration dosage and radiation dosage. Social and ethical implications of biological weapons

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : The students shall get an adequate knowledge on patent, Trademark and copyright. This provides further way for developing their idea or innovations
CO2 : Identify the role of significance of patent rights
CO3 : Students should get enough information on ethical issues linked to research on animal models, transgenic, clinical trials
CO4 : Get acquainted with Intellectual Property (IP) and Regulatory affairs as a career option.
CO5 : Understand the basics about biosafety
CO6 : Evaluate the effectiveness and limitations of computational chemistry methods and chemoinformatics tools in rational drug design

TEXT/REFERENCE BOOKS

1. Nithyananda, K V. (2019). “Intellectual Property Rights: Protection and Management”. India, IN: Cengage Learning India Private Limited.
2. Neeraj, P., & Khusdeep, D. (2014). “Intellectual Property Rights”. India, IN: PHI learning Private Limited.
3. V Sreekrishna, 2017. “Bioethics and Biosafety in Biotechnology” by New Age International publishers.
4. Goel D, “IPR, Bio safety and Bioethics”, Pearson Education, 2013.

24BT402T					Nanotechnology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	IA	ES	LW	LE/VIVA	
3	0	0	3	3	25	25	50	50	50	100

COURSE OBJECTIVES

1. To understand the fundamentals of Nanotechnology.
2. Explain the nanoscale paradigm in terms of properties at the nano scale dimension. Identify current nanotechnology solutions in design, engineering and manufacturing.
3. To understand and learn about the applications of nanoscale materials and the applicationof the principles of nano chemistry in real world problems.

UNIT I: INTRODUCTION 10 Hrs.

Introduction and History of Nanotechnology, Nano-materials and their properties: physical, chemicaland biological properties from their constituent atoms or molecules and from the bulk materials. Casestudies demonstrating non-classical behaviour at nanoscale in successful and emergent nanotechnologies.

UNIT II: SYNTHESIS OF NANO-MATERIALS 10 Hrs.

Synthesis of nano materials, bottom-up approach: self-assembly and self-organization, vapor phase deposition, plasma assisted deposition processes, colloidal, sol-gel, or simple pyrolysis, top-down approach: miniaturization of smaller structures from larger ones like milling, lithography, machining will be presented with suitable examples.

UNIT III: CHARACTERIZATION OF NANO-MATERIALS 10 Hrs.

Detailed characterization technique based on radiation matter interactions and their analytical applications like Transmission electron microscope (TEM), Scanning electron microscope (SEM), Atomic force microscope (AFM), scanning tunnelling microscope (STM), Dynamic light scattering (DLS), Spectroscopy. The safety and storage issues and the impact of nanotechnology on the environment will be stressed at the end.

UNIT IV: APPLICATIONS OF NANO-MATERIALS 10 Hrs.

Applications of Nanoscale materials: Catalysis, Sensing, Energy storage, Environmental remediation, etc.; Energy Conversion and Storage (solar cells, Fuel Cells, Li/Na-ion Batteries); Nanoscale materials for waste reduction and improved energy efficiency, nanotechnology-based water treatment strategies. Nano sensors and Devices.

42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Tell the basics of Nanotechnology along with properties
CO2 : Explain the chemical methods for synthesis of nanoparticles
CO3 : Develop various nanomaterials and basic understanding in the relevant analytical techniques
CO4 : Categorize the various techniques for nano-materials characterization
CO5 : Explain the physical methods for synthesis of nanoparticles
CO6 : Discuss the different applications of nano scale materials

TEXT/REFERENCE BOOKS

1. Cao G ,"**Nanostructures and Nanomaterials: Synthesis, Properties and Applications**" 2011, Imperial College Press.
2. Reich S and Maultzsch J, "**Carbon Nanotubes – Basic Concepts and Physical Properties**" 2008, Wiley.
3. Chattopadhyay K K ,"**Introduction to Nanoscience and Nanotechnology**", 2009, Prentice Hall India Learning Private Limited.
4. Sulabha K Kulkarni ,"**Nanotechnology: Principles and Practices**", 2014, Springer International Publishing

24BT402P	Nanotechnology lab

Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. To understand physical methods for synthesis of nanoparticles.
2. To understand the limitations of the synthesis techniques.
3. To understand various advancement in Techniques for synthesis at nanoscale.
4. To understand the use of nanotechnology in Energy Storage.

LIST OF EXPERIMENTS

1 Ball milling route for making nanoparticles and particle size distribution estimation.
2 Sol-gel synthesis and characterization of CdS nanocrystals
3 Preparation and characterization of ZnO nanoparticles embedded in silica matrix
4 Microwave-assisted synthesis of ZnO nanoparticles
5 Eco-Friendly Bio-Chemical synthesis of nanomaterials
6 Synthesis and Characterization of carbon nanotubes by cracking of gas mixture
7 Physical vapor deposition and chemical vapor deposition techniques for thin film deposition
8 Fabrication of suitable structures on thin films for device applications.
9 To investigate refluxing and distillation techniques for the synthesis of II-VI ceramic nanostructures
10 To study the solvo thermal synthesis method of nanoparticles

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Explain the basics of nanotechnology along with its properties.
CO2 : Explain the chemical methods for the synthesis of nanoparticles.
CO3 : Develop various nanomaterials with relevant analytical techniques
CO4 : Categorize the various techniques for nanomaterial characterization
CO5 : Explain the physical methods for the synthesis of nanoparticles
CO6 : Discuss the use of nanotechnology in Energy Storage.

TEXT/REFERENCE BOOKS

1. Guozhong Cao and Ying Wang," Nanostructures and Nanomaterials: Synthesis, Properties and Applications" 2011, Imperial College Press.
2. Gabor L. Hornyak, H.F. Tibbals, Joydeep Dutta, John J. Moore," Introduction to Nanoscience and Nanotechnology" by, 2008, Wiley.

24BT403T	Downstream Processing

Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

1. To apply the fundamental concepts of bio-separation engineering.
2. To learn and design downstream processing for product isolation and purification.
3. To recognize and troubleshoot problems associated with purification of bio products.
4. To integrate Advanced Bio-separation Techniques: Explore cutting-edge methods and technologies in downstream processing for efficient and sustainable recovery of bio-products at an industrial scale.

UNIT I: INTRODUCTION TO SCHEMATICS OF DOWNSTREAM PROCESSING 10 Hrs.

Range and characteristics of bio products, Characteristics of Fermentation Broth, Selection of unit operation with due consideration of physical, chemical and biochemical aspect of biomolecules, Stages of Downstream Processing. Cell disruption techniques: Mechanical and Non-mechanical methods of cell disruption, Mechanical Cell disruption methods Cell disruption kinetics for a bead mill and high-pressure homogenizer.

UNIT II: SEPARATION 10 Hrs.

Flocculation and sedimentation, Centrifugation-differential and density gradient. Filtration: incompressible and compressible cakes, Pre-treatment of fermentation broth, Filter media and equipment for batch and continuous Filtration. Membrane separation processes: Basic principles and advantages, Modes of operation, Pressure-driven processes (MF, UF, NF & RO), Concentration-driven processes (Dialysis &Pervaporation), and Electrically-driven processes (Electro dialysis); Applications of the different types of membrane separation processes. Chromatographic separation processes.

UNIT III: EXTRACTION AND PRECIPITATION 10 Hrs.

Extraction processes - solvent extraction principles, operating modes of extraction, Extraction of biomolecules by aqueous –aqueous, aqueous two-phase, reverse micellar, and supercritical fluid extraction. Colloidal stability of protein solutions, precipitation of biomolecules by isoelectric, salting out, organic solvent, non-ionic polymers, Poly- electrolyte addition and selective denaturation of unwanted proteins.

UNIT IV: FINISHING OPERATIONS 10 Hrs.

Crystallization, Drying, Lyophilization and Formulation, Industrial-scale Crystallizers and Dryers

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Assess the significance of downstream processing in a bio- product separation
CO2 : Apply the knowledge of unit operations for the separation of insoluble from fermentation broth
CO3 : Propose and design recovery and purification of biomolecules by applying concepts of mass transfer operation.
CO4 : Examine skills needed to function in modern bio separation engineering
CO5 : Design and demonstrate adsorption and chromatography processes for the purification of bio molecules.
CO6 : Demonstrate the skills and techniques to design a process for product purification.

TEXT/REFERENCE BOOKS

1. Bioseparations - Principles and techniques, B. Sivasankar, Prentice Hall of India, N Delhi, 2005, pp 280
2. Product recovery in Bioprocess technology (1992) Butterworth- Heinemann, Biotol series.
3. Mukesh D, Gaikar V and Anil Kumar Biotransformation's& Bioprocesses, Marcell Dekker, New york,(2004).

24BT403P	Downstream processing lab

Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Learn Basic Extraction and Purification of Biomolecules
2. Implement Diverse Separation Strategies
3. Extract and Isolate Natural Pigments
4. Employ Adsorption Techniques for Sample Enrichment

LIST OF EXPERIMENTS

1 Partial purification of intracellular/ extracellular products from microbial /plant sources—Extraction of intracellular metabolites by cell lysis.

2 Concentrating the intracellular extract by membrane filtration.

3 Separation of proteins based on the solubility by precipitation using salt.

4 Removal of salt from the protein solution and buffer exchange by dialysis

5 Isolation of carbohydrate digesting enzymes from cereal kernels – Drying, milling, and solid-liquid (aqueous) extraction

6 Separation of proteins using liquid-liquid extraction – Aqueous two-phase / reverse micellar process

7 Separation of metabolites based on molecular mass-gel filtration

8 Separation of pigments from selected flowers -Use of organic solvent extraction for separating the metabolites

9 Use of adsorption as the sample enrichment aid

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1: Demonstrate the ability to extract and partially purify biomolecules from microbial or plant sources.
CO2: Apply membrane filtration and salt precipitation techniques for protein concentration and separation.
CO3: Perform dialysis for effective salt removal and buffer exchange in protein solutions.
CO4: Isolate carbohydrate-digesting enzymes from cereal kernels using solid-liquid extraction methods.
CO5: Employ advanced liquid-liquid extraction methods, including aqueous two-phase and reverse micellar processes.
CO6: Utilize gel filtration, organic solvent extraction, and adsorption techniques for metabolite separation and enrichment.

TEXT/REFERENCE BOOKS

1. Bioseparations - Principles and techniques, B. Sivasankar, Prentice Hall of India, N Delhi, 2005,

24PRBT451					Seminars					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	-	-	-

COURSE OBJECTIVES

1. Expose students to cutting-edge technologies in biotechnology by engaging with industry experts and subject matter specialists.
2. Bridge the gap between academia and industry by discussing commercialization strategies, research gaps, and industrial expectations.
3. Enhance student networking opportunities with industry professionals, fostering collaborations and potential career guidance.
4. Develop critical thinking and presentation skills by encouraging student-led discussions, research reviews, and participation in technical debates.

Suggested Topics for Expert Talks and Panel Discussions:

1. Biopharmaceuticals and Drug Development
   - Emerging Trends in Biologics and Biosimilars
   - Vaccine Development: Challenges and Innovations
   - Monoclonal Antibodies and Their Therapeutic Applications
2. Targeted Drug Delivery Systems & Nanotechnology in Medicine
   - Role of Liposomes, Micelles, and Nano-carriers in Drug Delivery
   - Advances in CRISPR-based Drug Targeting
   - Nanomedicine for Cancer Therapy
3. Antimicrobial Resistance (AMR) and Future Therapeutics
   - Antibiotic Resistance Mechanisms and Global Challenges
   - Phage Therapy and Alternative Antimicrobial Strategies
   - Role of AI and In-Silico Approaches in Antimicrobial Drug Discovery
4. Gene Therapy and Genomic Engineering
   - CRISPR-Cas9 and its Biomedical Applications
   - Gene Editing for Rare Genetic Disorders
   - Ethical Considerations in Gene Therapy
5. Cancer Biology and Precision Medicine
   - Onco-Genomics and Personalized Cancer Therapy
   - Targeting Tumor Microenvironment in Cancer Treatment
   - Biomarkers in Early Cancer Diagnosis
6. In-Silico Drug Discovery and Bioinformatics Applications
   - Role of Computational Biology in Drug Development
   - AI and Machine Learning in Biopharmaceutical Research
   - Molecular Docking and Simulation Techniques for Drug Screening
7. Industrial Biotechnology and Commercialization Strategies
   - Scaling Up Biotechnology Processes: Challenges and Solutions
   - Biotech Startups and Entrepreneurship: Road to Commercialization
   - Regulatory Frameworks and Patent Strategies in Biotechnology
8. Emerging Fields in Biotechnology
   - Synthetic Biology and Its Industrial Applications
   - 3D Bioprinting for Tissue Engineering
   - Neurobiotechnology and Brain-Computer Interfaces

COURSE OUTCOMES

On completion of the seminars, students will be able to

CO1: Analyze the latest advancements in biopharmaceuticals, nanotechnology, gene therapy, and related biotechnological innovations.

CO2: Identify industry requirements and research gaps in biotechnology-related fields such as antimicrobial resistance, personalized medicine, and synthetic biology.

CO3: Demonstrate an understanding of the commercialization process, regulatory challenges, and intellectual property rights in biotechnology.

CO4: Engage effectively in scientific discussions and networking with industry professionals, enhancing career prospects.

CO5: Develop and deliver professional presentations on biotechnology-related topics, improving communication and technical articulation skills.

CO6: Propose innovative solutions and research ideas based on insights gained from industry experts and subject matter specialists.